

E.8 DETAILS

- **-1: No details to compare** - Neither outline provides details
- **0: Completely different** - Details differ entirely (e.g., benefits vs. generic notes)
- **1: Minimal depth** - Very limited overlap (e.g., one shared concept only)
- **2: Moderate depth** - Shared details plus some supporting facts
- **3: Highly specific details** - Quantitative or precise overlaps (e.g., percentages, statistics)

E.9 CONCISENESS OF HEADINGS

- **0: Verbose** - Headings too wordy or sentence-like
- **1: Mixed clarity** - Some concise, others verbose
- **2: Mostly concise** - Headings generally outline-appropriate

E.10 IDENTICAL ASSESSMENT

- **0: Not identical** - Outlines differ in content, even if similar
- **1: Identical** - Outlines essentially the same (ignoring trivial formatting)

Original Outline Generate	TAE Outline Residual Stream Decoder
Outline: 1. Quantum Uncertainty Principle - Limits measurement precision 2. Implications of the principle - Challenges classical determinism 3. Quantum mechanics nature - Probabilistic and uncertain 4. Measurement impact - Collapses wave function 5. Fundamental concept	Summary: 1. Quantum involution principle - Uncertainty of the quantum work 2. Relativity - Uncertainty of the quantum work 3. Implications of boundedness - Theoretical estimation 4. Implications of the uncertainty of the quantum work 5. Implications of the quantum work 3. Implications of the uncertainty of the quantum work 4. Implications of the quantum work 5.

Table 2: Comparison of Original Outline Generate vs TAE Outline Residual Stream Decoder - Example 5

F TEMPORAL DYNAMICS: TOKEN REPLACEMENT ANALYSIS

Given that the model does not seem to form plans for the next paragraph, we attempt to further understand what causes the model to plan for the next paragraph. We perform the same procedure as we did in Section 6.2 but when generating the residual stream, replace the input token with "\n\n" to see if the specific token is responsible for the model generating the plans.

We conduct detailed analysis of how the model handles artificially inserted paragraph transitions. For each text pair, we:

- 1) Take original text with natural "\n\n" transition
- 2) Replace tokens at positions $[-10, +10]$ with "\n\n"
- 3) Apply Continuation ParaScope
- 4) Compare outputs with original second paragraph

Results show position-dependent effects: - Peak similarity at original transition (0.62 ± 0.14) - Gradual decline pre-transition (0.45 ± 0.12 at -5 tokens) - Steeper decline post-transition (0.31 ± 0.13 at +5 tokens) - High variance in token-specific effects ($\sigma = 0.18$)

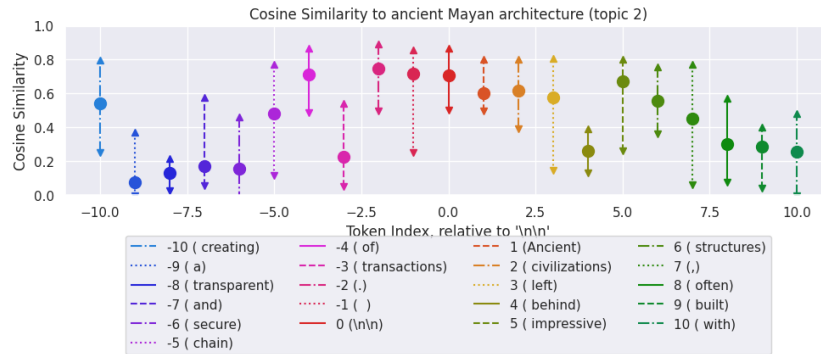


Figure 13: explanation of how we do token-wise analysis with manipulation to replace `\n\n` to see when the LLM prominently seems to be looking forward

G SONAR LINEAR MAP

G.1 SONAR PARASCOPE LINEAR MAP DETAILS

We train the model on a subset of 100,000 samples, taking the normalized residual stream activations as the input, and SONAR embeds of the paragraphs as the output.

Activations were normalized by using Welford’s Online Algorithm on the first 10,000 texts (approx 100,000 samples) to compute mean and standard deviations, and using this to achieve approximately unit normal distribution on each dimension.

We train only on the output of the MLP and Attention (residual diff) of the final 12 layers (giving a total of 24 sub-layers) from Llama 3.2 3B’s model’s total of 28 layers.

In principle, one could use any text auto-encoder, or use a more complex probe such as an MLP probe.

Training hyperparameters were as follows:

- Linear map:
- Batch size: 1,024
- Weight decay: $1e-7$
- Learning rate: $2e-5$ with $\times 0.8$ decay/epoch
- Epochs: 10

G.2 SONAR LINEAR MAP ANALYSIS

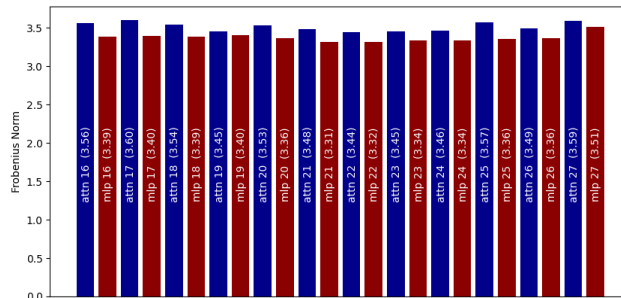


Figure 14: Layerwise Frobenius norm of Linear Regression Map from Sonar Map ParaScope

We perform detailed analysis of the learned SONAR mappings:

Layer-wise Weight Distribution: - Computed per-layer Frobenius norms - Attention layers: mean norm 3.56 (± 0.42) - MLP layers: mean norm 3.39 (± 0.38) - Layer-wise correlation $\tau = 0.73$

G.3 CORRELATION ANALYSIS

We examine inter-method correlations using Kendall’s τ , including preliminary results with an MLP-based probe³.

Between Methods: - MLP vs Linear: $\tau = 0.82$ - MLP vs Continuation: $\tau = 0.41$ - Linear vs Continuation: $\tau = 0.43$

Between Metrics (Linear ParaScope): - Length vs Subject: $\tau = 0.58$ - Coherence vs Detail: $\tau = 0.45$ - Entity vs Subject: $\tau = 0.61$

Between Metrics (Continuation): - Length vs Subject: $\tau = 0.39$ - Coherence vs Detail: $\tau = 0.31$ - Entity vs Subject: $\tau = 0.44$

These results suggest SONAR variants learn similar mappings but capture different aspects than Continuation ParaScope.

H USAGE OF LLMs

We use LLMs at various parts of the process.

We use tools such as ChatGPT, Claude, Github Copilot, and Cursor when writing code and running experiments, and were iteratively used to create rubrics for scoring outputs, though all code is ultimately human-reviewed.

For the literature review process of finding papers, tools like Google Scholar, Semantic Scholar, Connected Papers, we used, with supplemental search often done by ChatGPT, Claude, and Perplexity.

When it comes to paper writing, spell checker tools and LLMs were used to sometimes check for writing clarity and sometimes suggest edits that were accepted.

I EXAMPLES OF PARASCOPE OUTPUTS

We show a randomly selected set of 5 examples of TAE ParaScope and Continuation ParaScope examples, with the reference generation side by side.

TAE	Cont.	Original
Understanding your migraine disorders is a key to managing and preventing migraine symptoms. By understanding your symptoms and preventing them, you can identify the best ways to manage and prevent migraine symptoms. By learning about your symptoms and preventing them, you can help manage your symptoms and prevent them.	Understanding the importance of sleep and establishing a consistent sleep schedule can have a significant impact on overall health and well-being. By making sleep a priority and creating a sleep-conducive environment, individuals can improve the quality of their sleep and wake up feeling refreshed and revitalized. Whether you’re looking to improve your sleep hygiene or simply need some tips to help you get a good night’s sleep, there are many resources available to help. So, take the first step towards better sleep and start making it a priority today.	Understanding your migraine triggers is the first step towards managing your symptoms and improving your overall quality of life. By exploring your individual triggers and taking action to manage them, you can reduce the frequency and severity of your migraines and take control of your health.

Table 3: Comparison of TAE vs Cont. vs Original - Example 1

³This includes preliminary results where we trained an MLP Sonar ParaScope probe with 8192 hidden layers. We did not include MLP results these in the final paper as the results were not appreciably better than with the Linear model, and highly correlated with the Linear model.

TAE	Cont.	Original
The CDF International Conference on Poetry 2019 was a prestigious event held in Cologne, Germany, that brought together the world’s leading scholars in poetry and literature. The event celebrated the 35th anniversary of the conference, which featured a comprehensive presentation of poetry and literature from across the globe. The conference was designed to bring together scholars from different backgrounds and disciplines. The conference featured a wide range of exhibitions on poetry, poetry, and literature.	The [Year] conference, held on [Date] at [Location], was a groundbreaking event that brought together leading experts in the field of [Field] to discuss the latest advancements and challenges in [Specific Area of Interest]. The conference featured a diverse range of speakers, including [Notable Speakers], who shared their insights and experiences on topics such as [Key Topics]. The event was well-attended by [Number] of delegates, who engaged in lively discussions and networking opportunities.	The 2015 CDI Poetry Conference was a premier event for poets and literature enthusiasts, held in the vibrant city of Toronto, Canada, from May 13th to 16th, 2015. The conference brought together over 200 poets, writers, and scholars from across Canada and around the world to share their work, engage in lively discussions, and explore the latest trends and themes in Canadian poetry. The conference featured an impressive lineup of notable attendees, including poet and Nobel laureate Margaret Atwood, critically acclaimed poet and essayist Tanya Talagant, and renowned publisher and literary critic Greg Gellenbeck.

Table 4: Comparison of TAE vs Cont. vs Original - Example 2

TAE	Cont.	Original
In the meantime, the new rules on investing are likely to lead to greater transparency and transparency in the decision-making process.	Overall, the recent developments in the field of quantum computing and the emergence of new technologies have created a new landscape for the development of quantum algorithms and applications. As researchers continue to push the boundaries of what is possible with quantum computing, we can expect to see new breakthroughs and innovations in the years to come.	In the meantime, startups and investors are already taking steps to prepare for the new rules. Many crowdfunding platforms are offering educational resources and training programs to help investors better understand the risks and benefits of investing in startups.

Table 5: Comparison of TAE vs Cont. vs Original - Example 3

J EXAMPLES OF TAE OUTLINE RSD OUTPUTS

We show a randomly selected set of y5 examples of TAE Outline Residual Stream Decoder outputs.